

# Operations Manual

How to prepare, run and recover a Boaty session safely: procedures, checklist, contingency cards and the crew card

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Document | BOATY-OPS-001                                                                     |
| Issue    | Issue E (baseline candidate, SDR)                                                 |
| Date     | 30 September 2026                                                                 |
| Status   | Baseline candidate for the SDR freeze; the owner is also its main user            |
| Basis    | SRS Issue H (OPS-001 to OPS-012, ConOps, ConUse); FMEA Issue G procedural actions |
| Content  | 14 procedures, 18-item checklist, 12 contingency cards                            |

# Document control

| Issue | Date              | Change                                                                                                                                                                         | By                             |
|-------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| A     | 28 September 2026 | First issue.                                                                                                                                                                   | Claude (drafted)               |
| B     | 30 September 2026 | Simulator findings: K-03 (a dead motor looks like weed), K-05 (the boat stops itself 30 s or 10 m outside the fence), K-11 (a heading alarm in strong wind may be wind drift). | Claude, from simulator slice 2 |
| C     | 29 September 2026 | Mission Control findings: CL-13 (arm at the jetty: home is taken where the boat is armed) and CL-18 (close any other ground station) added.                                    | Claude, from simulator slice 3 |
| D     | 30 September 2026 | SDR decision CR-07: site drawing now puts a $\geq 15$ m circle round every nest structure; bird photos only from outside it (OPS-005, SRS Issue G).                            | Claude, owner decision         |
| E     | 30 September 2026 | SDR baseline candidate: OP-12 step on session logs (WP4, RID-14); basis from the baseline register (RID-08).                                                                   | Claude                         |

| Role                | Name                      | Signature / date |
|---------------------|---------------------------|------------------|
| Author              | Claude, for Project Boaty |                  |
| Reviewer / approver | Project owner             |                  |

Review guidance: read it as if it's your first lake session. Anything you wouldn't actually do, or that's missing, is a finding. Sections 7 and 8 are meant to be printed and laminated.

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# 1. About this manual

This is the manual for using Boaty, from the day before a session to putting the battery away. It turns the operations requirements (OPS-001 to OPS-012) and the procedural FMEA actions into things to do, in order. It assumes the boat and software meet their specifications. Where software helps (the checklist, alarms, spoken messages), the procedure says so.

- **Section 5, the pre-launch checklist**, is the same list Mission Control shows before it allows arming (MCN-D13). Edit it here, and both change.
- **References** in brackets (e.g. OPS-004, A-17) point to the requirement or FMEA action each step satisfies. That tells you why a step exists before you're tempted to skip it.
- **Sections 7 and 8** (crew card, quick reference) are one page each, for printing.

# 2. Roles and golden rules

| Role             | Responsibilities                                                                                                                                                    |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operator (adult) | In charge of safety from packing to unpacking. Holds the magnetic arming key and the PIN. Is the lookout during every mission. Makes every go / no-go call.         |
| Crew (age 4)     | Builds and decorates the boat, gives the orders with TALK, presses GO, COME HOME and STOP, spots the boat, finds the ducks. Never goes near the water's edge alone. |

**1. People first, boat second.** If anyone could get hurt, forget the boat. It floats and it can wait. (OPS-007)

**2. Nobody goes in the water.** Not for the boat, not for anything. Use the pole, the net or the casting line. (OPS-008)

**3. Key out before hands on.** Pull the magnetic key before anyone touches the boat, and keep it until the boat is on the water. (A-09)

**4. The operator is the lookout.** Eyes on the boat and the water for the whole mission. If in doubt, STOP. (A-17, OPS-004)

**5. Never on shared water without permission.** Milton means the Trust's permission and its conditions. (OPS-002)

### 3. Kit lists

|                          | Group        | Item                                                                                      |
|--------------------------|--------------|-------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Boat         | Boat (hulls, pods fitted, mast, flag)                                                     |
| <input type="checkbox"/> | Boat         | DUPLO crew                                                                                |
| <input type="checkbox"/> | Boat         | Magnetic arming key on its lanyard (operator's pocket)                                    |
| <input type="checkbox"/> | Bank station | Mission Control box                                                                       |
| <input type="checkbox"/> | Bank station | USB-C power bank (charged)                                                                |
| <input type="checkbox"/> | Bank station | Phone + USB cable (tether and screen)                                                     |
| <input type="checkbox"/> | Bank station | Antenna pole kit (if V-08 showed it's needed)                                             |
| <input type="checkbox"/> | Battery      | Boat battery in a fire-safe bag, charged the day before, carried separately from the boat |
| <input type="checkbox"/> | Recovery kit | Telescopic pole $\geq 3$ m with landing net                                               |
| <input type="checkbox"/> | Recovery kit | Casting rod with weighted float line                                                      |
| <input type="checkbox"/> | Recovery kit | Throw towel                                                                               |
| <input type="checkbox"/> | Spares kit   | Spare props                                                                               |
| <input type="checkbox"/> | Spares kit   | Spare 20 A fuse                                                                           |
| <input type="checkbox"/> | Spares kit   | Cable ties                                                                                |
| <input type="checkbox"/> | Spares kit   | Microfibre lens cloth                                                                     |
| <input type="checkbox"/> | Spares kit   | Towel                                                                                     |
| <input type="checkbox"/> | Spares kit   | Small screwdriver (adult use)                                                             |
| <input type="checkbox"/> | People       | Drinks and snacks                                                                         |
| <input type="checkbox"/> | People       | Hand gel or wipes (algae)                                                                 |
| <input type="checkbox"/> | People       | Sun hats / waterproofs                                                                    |

## 4. Procedures

### OP-01 Setting up a new site (once per site)

| When                                                                                                          | Before the first session at any water                                                                                                                                                                                                                                                               | Who              | Operator |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------|
| Make sure the fence matches the real water, and that we're allowed and welcome there.                         |                                                                                                                                                                                                                                                                                                     |                  |          |
| #                                                                                                             | Step                                                                                                                                                                                                                                                                                                | Why              |          |
| 1                                                                                                             | Get written permission from the landowner. At Milton, email Cambridge Sport Lakes Trust and record their conditions in the site notes.                                                                                                                                                              | OPS-002          |          |
| 2                                                                                                             | Draw the site on the map at home: inclusion fence 5 m inside the waterline, home points, areas and landmarks.                                                                                                                                                                                       | FEN-002, FEN-003 |          |
| 3                                                                                                             | Add exclusion zones over islands, reed beds and angling swims ( $\geq 20$ m clearance). Put a circle of at least 15 m radius round every nest structure (duck house, nest raft) and tag it wildlife = nest; the site check refuses less. Photos of birds are taken only from outside these circles. | OPS-004, OPS-005 |          |
| 4                                                                                                             | Keep the first fence within 100 m of the launch point. Grow it only after 5 successful missions logged at this site.                                                                                                                                                                                | OPS-006          |          |
| 5                                                                                                             | <b>Fence survey walk.</b> On the first visit, walk the launch bank carrying the boat's GNSS (or the phone) with logging on. Compare the logged track with the drawn waterline. If they differ by more than 3 m, shift the fence and record the offset in the site file.                             | A-05             |          |
| 6                                                                                                             | Mark launch points and the wind directions each suits (wind blowing towards that bank).                                                                                                                                                                                                             | A-19             |          |
| 7                                                                                                             | Run the site linter. It must pass before the site can be used.                                                                                                                                                                                                                                      | MCN-D53          |          |
| Redo the survey walk if the water level has changed a lot or the boat ever reaches the bank inside its fence. |                                                                                                                                                                                                                                                                                                     |                  |          |

### OP-02 Trial progression

| When                                                               | Once, from first power-up to first lake mission                                           | Who              | Operator |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------|----------|
| Never find a problem on the lake that we could have found at home. |                                                                                           |                  |          |
| #                                                                  | Step                                                                                      | Why              |          |
| 1                                                                  | Simulation: every failsafe scenario passes (SC-01 to SC-13), and the FMEA scenarios run.  | OPS-009, SWE-005 |          |
| 2                                                                  | Bench rigs L1 and L2: STOP, key, ESC signal loss and brownout tests pass.                 | OPS-009          |          |
| 3                                                                  | Paddling pool: floats level, drives, stops, stuck detection, photo sync, crew controls.   | OPS-009          |          |
| 4                                                                  | Small private water (if available): first autonomous missions within casting range.       | OPS-009          |          |
| 5                                                                  | Lake: only when every SIM, BENCH and POOL Must requirement has passed, and OP-03 is done. | OPS-009, OPS-012 |          |

### OP-03 Rehearsal at home

| When                                                | Before the first lake trial, then each spring                                              | Who     | Operator (and crew) |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------|---------|---------------------|
| Practise every contingency where mistakes are free. |                                                                                            |         |                     |
| #                                                   | Step                                                                                       | Why     |                     |
| 1                                                   | Start the simulator in rehearsal mode with the real panel.                                 | SIM-D05 |                     |
| 2                                                   | Work through every contingency card (section 6) at least once: recognise it, act, recover. | OPS-012 |                     |
| 3                                                   | Let the crew practise TALK, GO, COME HOME and STOP on the virtual lake.                    | OPS-012 |                     |

### OP-04 The day before

|                                                                      |               |     |          |
|----------------------------------------------------------------------|---------------|-----|----------|
| When                                                                 | Every session | Who | Operator |
| Arrive with a healthy battery, the right software and the right kit. |               |     |          |

| # | Step                                                                                                                                                             | Why              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1 | <b>Battery inspection:</b> no swelling, dents, wrapper damage or smell. Cell voltages balanced within 0.05 V. Pack at room temperature. Any doubt: don't use it. | A-11, PWR-D20    |
| 2 | Charge on a balance charger, in the fire-safe bag, on a non-flammable surface, attended. Never charge in the boat.                                               | OPS-010, PWR-007 |
| 3 | Software: use the tagged release only. Run the simulator suite if anything changed.                                                                              | SWE-006          |
| 4 | Check the forecast: wind at the lake $\leq$ Beaufort 3, no storms.                                                                                               | OPS-001          |
| 5 | Pack the kit lists (section 3), including the recovery and spares kits.                                                                                          | OPS-008, OPS-011 |

## OP-05 Transport

|      |               |     |          |
|------|---------------|-----|----------|
| When | Every session | Who | Operator |
|------|---------------|-----|----------|

Arrive with nothing broken or live.

| # | Step                                                      | Why           |
|---|-----------------------------------------------------------|---------------|
| 1 | Battery in its fire-safe bag, separate from the boat.     | PWR-007       |
| 2 | Magnetic key in the operator's pocket, never on the boat. | A-09, MOD-003 |
| 3 | Boat in its cradle with the mast removed.                 | MEC-014       |

## OP-06 Setting up on the bank

|      |               |     |                        |
|------|---------------|-----|------------------------|
| When | Every session | Who | Operator, crew helping |
|------|---------------|-----|------------------------|

Decide whether today is a go, and set up safely.

| # | Step                                                                                                                   | Why              |
|---|------------------------------------------------------------------------------------------------------------------------|------------------|
| 1 | Read the notice boards: no swim session, no watersports course, no algae warning. If any, go home and try another day. | OPS-003          |
| 2 | Look at the water: who else is here? Keep $\geq$ 20 m from anglers' lines and other users.                             | OPS-004          |
| 3 | Check the wind. Choose the launch point the site file suggests for today's wind, so a stopped boat drifts to us.       | A-19, OPS-001    |
| 4 | Set up Mission Control a few metres back from the edge. Plug in the phone and start the app.                           | OPS-007          |
| 5 | Crew fits the pods and the DUPLO crew. <b>Key stays in the operator's pocket.</b>                                      | A-09, CHD-004    |
| 6 | Fit the battery, close and latch the box, main switch on.                                                              | MEC-012          |
| 7 | Work through the pre-launch checklist on screen (section 5). Every item must be ticked before arming is possible.      | OPS-001, PRE-006 |

## OP-07 Crew briefing

|      |                                |     |                  |
|------|--------------------------------|-----|------------------|
| When | Every session, before planning | Who | Operator to crew |
|------|--------------------------------|-----|------------------|

Four buttons, four rules, in words a 4-year-old knows.

| # | Step                                                                                                                           | Why     |
|---|--------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 | Show the four buttons (see the crew card): blue TALK to tell the boat, green GO to start, yellow house to come home, red STOP. | CHD-005 |
| 2 | Rule 1: we stay behind the line (a rope or bag marks it).                                                                      | OPS-007 |
| 3 | Rule 2: only grown-ups touch the boat when it's on the water.                                                                  | A-09    |
| 4 | Rule 3: if you're worried, press the red button. Nobody will be cross.                                                         | FS-009  |
| 5 | Rule 4: we never go in the water, even for the boat.                                                                           | OPS-008 |

## OP-08 Planning the mission

When Every mission

Who Crew speaks, operator approves

Turn the crew's idea into a safe plan.

| # | Step                                                                                                                           | Why              |
|---|--------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1 | Crew holds TALK and says what the boat should do.                                                                              | NLI-002          |
| 2 | Listen to the summary and look at the map: duration, distance, photo count, furthest point from home.                          | MCN-D56, NLI-005 |
| 3 | Check the route stays well clear of other people and wildlife. If anyone is within 50 m of the route, change the plan or wait. | A-17, OPS-005    |
| 4 | Approve with the PIN (never read the PIN aloud). No internet: choose a template with the crew.                                 | VAL-008, MIS-005 |

## OP-09 Launch and arming

When Every mission

Who Operator

Get the boat on the water with nobody's hands near a prop.

| # | Step                                                                                         | Why            |
|---|----------------------------------------------------------------------------------------------|----------------|
| 1 | Carry the boat by the handle, put it on the water at the chosen launch point, and step back. | A-09           |
| 2 | Only now insert the magnetic key into its dock.                                              | A-09, MOD-003  |
| 3 | Arm at Mission Control. Check home is shown in the right place.                              | MOD-003, FM-46 |
| 4 | Crew holds GO for a second: 'Off we go!'                                                     | MC-004         |

## OP-10 During the mission: the lookout

When Whole mission

Who Operator

Be the eyes the boat doesn't have.

| # | Step                                                                                                                            | Why           |
|---|---------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1 | Watch the boat and the water around it all the time. Hand the phone to nobody.                                                  | A-17          |
| 2 | Anyone (swimmer, paddler, dog, angler's line) within 20 m of the boat or 50 m of its route: press COME HOME, or STOP if closer. | OPS-004, A-17 |
| 3 | Birds with young, or the boat heading for reeds or nests: COME HOME.                                                            | OPS-005       |
| 4 | Keep the crew behind the line. Being with the child comes before watching the boat.                                             | OPS-007       |
| 5 | Anything strange (circling, wrong direction, alarms): STOP first, think second (section 6).                                     | FM-17, FM-05  |

## OP-11 Recovering the boat

When End of every mission

Who Operator

Get the boat out with the motors dead.

| # | Step                                                                                                  | Why           |
|---|-------------------------------------------------------------------------------------------------------|---------------|
| 1 | Boat waits by the bank at home (HOLD). Disarm at Mission Control.                                     | MOD-005       |
| 2 | <b>Pull the magnetic key</b> and pocket it, then lift the boat by the handle or hoop. Never by a pod. | A-09          |
| 3 | Look at the guards and props: cracked guard = no more missions today.                                 | A-09, MEC-010 |

## OP-12 After the session

When Every session

Who Operator, crew for photos

Leave the boat, battery and records ready for next time.

| # | Step                                                                                                                                                                                                                                                      | Why              |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1 | Let photos sync and make the captain's log together.                                                                                                                                                                                                      | DET-002          |
| 2 | Rinse pods and hulls with clean water and dry the motors. Everyone washes hands.                                                                                                                                                                          | OPS-003          |
| 3 | Main switch off only after Mission Control says the mission computer has shut down.                                                                                                                                                                       | IF-08            |
| 4 | Battery out, home in the fire-safe bag. Within a day, storage-charge it ( $\approx 3.7$ V/cell).                                                                                                                                                          | OPS-010          |
| 5 | Download logs. Note anything odd as an incident (OP-14).                                                                                                                                                                                                  | LOG-003          |
| 6 | The session log holds what was said to Boaty. It stays on Mission Control and is never uploaded. Delete a session's folder when it is no longer useful (the only thing that ever left the bank was each instruction's text, to plan the trip; ICD IF-11). | LOG-002, LOG-004 |

## OP-13 Maintenance

| When                             | See table                                                                                                           | Who                       | Operator |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------|----------|
| Catch wear before it catches us. |                                                                                                                     |                           |          |
| Interval                         | Tasks                                                                                                               | Refs                      |          |
| Every session                    | Rinse and dry pods and motors; inspect guards; check box seal; battery inspection (OP-04)                           | A-09, A-11                |          |
| Monthly (in season)              | Replace worn props; check thumb-screws and latches; clean camera window; check the SD card health report            | MEC-007                   |          |
| Each season                      | Box dunk test; re-measure pack capacity (PWR-D21); change the PIN; re-run the full simulator suite; review the FMEA | PWR-D21, SAF-006, MCN-D61 |          |
| After any change                 | New firmware, parameters or code: tagged release, full simulator suite, rig retest of anything touched              | SWE-006, SAF-007          |          |

## OP-14 Incidents and learning

| When                              | Whenever something unexpected happens                                                            | Who              | Operator |
|-----------------------------------|--------------------------------------------------------------------------------------------------|------------------|----------|
| Every surprise makes Boaty safer. |                                                                                                  |                  |          |
| #                                 | Step                                                                                             | Why              |          |
| 1                                 | Write down what happened, when, and what the logs show.                                          | LOG-003          |          |
| 2                                 | Check it against the FMEA. New failure mode or worse than rated? Update the FMEA and add a test. | SAF-006          |          |
| 3                                 | Anything involving people or wildlife: pause lake sessions until it's understood.                | OPS-004, OPS-005 |          |

## 5. Pre-launch checklist

Every item must be ticked before Mission Control allows arming (PRE-006). 'Automatic' items are ticked by the system and can't be overridden. Record of who signed off and when is kept in the session log (MC-010).

| ID                             | Group           | Check                                                                                                                                                                        | How                             | Why                       |
|--------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------|
| <input type="checkbox"/> CL-01 | Site and people | Permission held and today's conditions met                                                                                                                                   | Operator                        | OPS-002                   |
| <input type="checkbox"/> CL-02 | Site and people | No swim session, watersports course or algae warning                                                                                                                         | Operator                        | OPS-003                   |
| <input type="checkbox"/> CL-03 | Site and people | Nobody within 50 m of the planned area; anglers' swims excluded                                                                                                              | Operator                        | OPS-004, A-17             |
| <input type="checkbox"/> CL-04 | Site and people | Wildlife exclusions reviewed on the map                                                                                                                                      | Operator                        | OPS-005                   |
| <input type="checkbox"/> CL-05 | Weather         | Wind Beaufort 3 or less (small waves, few white horses)                                                                                                                      | Operator                        | OPS-001                   |
| <input type="checkbox"/> CL-06 | Weather         | Wind blowing towards our bank; launch point chosen                                                                                                                           | Operator (suggested by the app) | A-19, OPS-001             |
| <input type="checkbox"/> CL-07 | Boat            | Battery inspected and $\geq 80\%$ charged                                                                                                                                    | Operator + automatic            | A-11, PRE-003             |
| <input type="checkbox"/> CL-08 | Boat            | Box sealed and latched; glands tight                                                                                                                                         | Operator                        | OPS-001, MEC-012          |
| <input type="checkbox"/> CL-09 | Boat            | Guards intact; props free; pods latched                                                                                                                                      | Operator                        | A-09, MEC-010             |
| <input type="checkbox"/> CL-10 | Boat            | Flag, hoop and beacon fitted and working                                                                                                                                     | Operator                        | OPS-001, REC-003          |
| <input type="checkbox"/> CL-11 | Boat            | Key-out rail test passed                                                                                                                                                     | Automatic                       | MCN-D54                   |
| <input type="checkbox"/> CL-12 | Bank            | Button test passed (TALK, GO, COME HOME, STOP)                                                                                                                               | Automatic                       | MCN-D55                   |
| <input type="checkbox"/> CL-13 | System          | Fence and exclusions reviewed; boat at the jetty when armed (the helm takes home from where it is armed; Mission Control refuses a plan more than 10 m from the site's home) | Operator                        | OPS-001, PRE-002, VAL-004 |
| <input type="checkbox"/> CL-14 | System          | Parameters match baseline; $\geq 8$ satellites; link good                                                                                                                    | Automatic                       | SAF-007, PRE-001          |
| <input type="checkbox"/> CL-15 | Recovery        | Recovery kit on the bank (pole and net, casting rod and line)                                                                                                                | Operator                        | OPS-008, OPS-001          |
| <input type="checkbox"/> CL-16 | Recovery        | Spares kit on the bank                                                                                                                                                       | Operator                        | OPS-011                   |
| <input type="checkbox"/> CL-17 | People          | Adult operator present and acting as lookout; crew briefed                                                                                                                   | Operator                        | OPS-007, A-17             |
| <input type="checkbox"/> CL-18 | System          | Anything else talking to the boat closed (QGroundControl, a second tablet): Mission Control refuses to arm while it hears one                                                | Operator                        | MCN-D57                   |

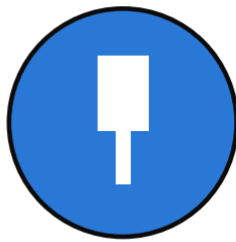
## 6. Contingency cards

What to do when something goes wrong. The first move is almost always the same: **people first, then STOP or COME HOME, then think.**

| K-01 Link lost                      |                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| You'll notice                       | 'Link lost' on screen and spoken; last position frozen on the map                                                                                                                                                              |
| Operator                            | Stay put or move to a better spot. In a mission, the boat carries on, then comes home by itself after 60 s. Driving manually, it stops, then comes home after 10 s.                                                            |
| Crew                                | Watch for the flag                                                                                                                                                                                                             |
| Refs                                | FS-002, FS-003                                                                                                                                                                                                                 |
| K-02 Low battery                    |                                                                                                                                                                                                                                |
| You'll notice                       | 'Coming home: battery low'                                                                                                                                                                                                     |
| Operator                            | Let it come home. End the session.                                                                                                                                                                                             |
| Crew                                | Wave it in                                                                                                                                                                                                                     |
| Refs                                | FS-001                                                                                                                                                                                                                         |
| K-03 Stuck in weed                  |                                                                                                                                                                                                                                |
| You'll notice                       | 'I'm stuck, trying to wiggle free'                                                                                                                                                                                             |
| Operator                            | Wait for up to three wiggles. Still stuck: COME HOME to retry, else recover by line (K-07). Stuck in clear water with no weed in sight? Suspect a dead motor: the boat can't tell the two apart. Recover and inspect the pods. |
| Crew                                | Spot the boat                                                                                                                                                                                                                  |
| Refs                                | FS-005, FS-006, FM-50                                                                                                                                                                                                          |
| K-04 Boat stopped: position lost    |                                                                                                                                                                                                                                |
| You'll notice                       | 'I've lost my way, stopping'                                                                                                                                                                                                   |
| Operator                            | Wait up to a minute. It goes home once GPS is back. If not, recover (K-07).                                                                                                                                                    |
| Crew                                | -                                                                                                                                                                                                                              |
| Refs                                | FS-004                                                                                                                                                                                                                         |
| K-05 At or beyond the fence         |                                                                                                                                                                                                                                |
| You'll notice                       | Fence alarm; boat heading home                                                                                                                                                                                                 |
| Operator                            | Let it return. If wind keeps it outside, it stops itself after 30 s or 10 m out ('outside fence' alarm): recover it (K-07). If it's pushed against the bank, STOP and recover.                                                 |
| Crew                                | -                                                                                                                                                                                                                              |
| Refs                                | FEN-005, FEN-006, MCP-D30                                                                                                                                                                                                      |
| K-06 Person or animal near the boat |                                                                                                                                                                                                                                |
| You'll notice                       | You see them first: the boat can't                                                                                                                                                                                             |
| Operator                            | COME HOME, or STOP if close. Wait until clear.                                                                                                                                                                                 |
| Crew                                | Press STOP if asked                                                                                                                                                                                                            |
| Refs                                | OPS-004, A-17                                                                                                                                                                                                                  |

| K-07 Boat dead in the water                    |                                                                                                                                                                                                                            |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| You'll notice                                  | Stopped; no response; beacon may be dark                                                                                                                                                                                   |
| Operator                                       | It floats. Wait for the wind to bring it to our bank. Pole and net if within reach. Otherwise cast the weighted line beyond the boat and reel it in so the line catches the mast and hoop. <b>Never wade.</b>              |
| Crew                                           | Stay behind the line                                                                                                                                                                                                       |
| Refs                                           | OPS-008, REC-005                                                                                                                                                                                                           |
| K-08 No internet                               |                                                                                                                                                                                                                            |
| You'll notice                                  | 'Templates only'                                                                                                                                                                                                           |
| Operator                                       | Choose a template with the crew.                                                                                                                                                                                           |
| Crew                                           | Pick an adventure                                                                                                                                                                                                          |
| Refs                                           | MIS-005                                                                                                                                                                                                                    |
| K-09 Mission Control dead                      |                                                                                                                                                                                                                            |
| You'll notice                                  | Screen and buttons dead                                                                                                                                                                                                    |
| Operator                                       | The boat comes home by itself after 60 s. Restart the box. Recover as normal.                                                                                                                                              |
| Crew                                           | -                                                                                                                                                                                                                          |
| Refs                                           | FS-003                                                                                                                                                                                                                     |
| K-10 Water or heat in the box                  |                                                                                                                                                                                                                            |
| You'll notice                                  | 'Water in the box' or 'too hot' alarm; boat returning                                                                                                                                                                      |
| Operator                                       | Let it return, key out, battery out, dry or cool, and investigate before the next mission.                                                                                                                                 |
| Crew                                           | -                                                                                                                                                                                                                          |
| Refs                                           | FS-010, A-11                                                                                                                                                                                                               |
| K-11 Boat behaving strangely                   |                                                                                                                                                                                                                            |
| You'll notice                                  | Circling, going the wrong way, or a navigation alarm                                                                                                                                                                       |
| Operator                                       | STOP. Look before resuming: pods, weed, compass. A 'heading check' alarm in strong wind may be the wind blowing the boat backwards, not the compass: check the wind first. Resume only with the PIN if you understand why. |
| Crew                                           | Press STOP if asked                                                                                                                                                                                                        |
| Refs                                           | FM-17, FM-05, FM-49, MCN-D60                                                                                                                                                                                               |
| K-12 Battery hot, swollen, smoking or smelling |                                                                                                                                                                                                                            |
| You'll notice                                  | Heat, hissing, smoke                                                                                                                                                                                                       |
| Operator                                       | Keep everyone away, especially the crew. Don't touch it. If it's on land and burning, let it burn out on a non-flammable surface or cool it with lots of water from a distance. Call 999 if fire spreads.                  |
| Crew                                           | Go with a grown-up, away from it                                                                                                                                                                                           |
| Refs                                           | FM-23, OPS-010                                                                                                                                                                                                             |

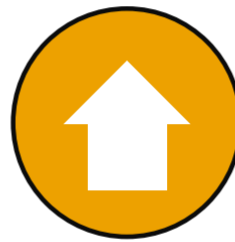
## 7. Crew card (print me)



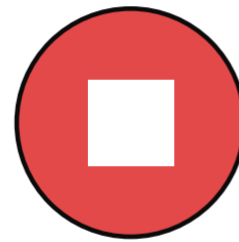
**TALK**



**GO**



**COME HOME**



**STOP**

Stay behind the line · Only grown-ups touch the boat · Never go in the water

| Button    | Looks like       | What it does                                   |
|-----------|------------------|------------------------------------------------|
| TALK      | Blue, microphone | Hold it and tell Boaty what to do.             |
| GO        | Green, arrow     | Hold it to start. Only when the grown-up says. |
| COME HOME | Yellow, house    | Boaty comes back to us.                        |
| STOP      | Red, square      | Boaty stops. Press it any time you're worried. |

Read it together before every session (OP-07). The crew can't break anything by pressing a button: GO only works when the grown-up has made it safe.

## 8. Quick reference (print me)

### Golden rules

1. People first, boat second.
2. Nobody goes in the water.
3. Key out before hands on.
4. The operator is the lookout.
5. Never on shared water without permission.

### A session in eight lines

OP-04 Day before: inspect and charge battery, pack kits

OP-06 Bank: notices, other users, wind → launch point

Checklist (section 5) → OP-07 crew briefing

OP-08 TALK → check plan → approve with PIN

OP-09 Boat on water → step back → KEY IN → arm → GO

OP-10 Lookout the whole time → STOP if in doubt

OP-11 Home → disarm → KEY OUT → lift by handle

OP-12 Sync photos, rinse, wash hands, storage-charge

| If...                                     | Do                                                                                          |
|-------------------------------------------|---------------------------------------------------------------------------------------------|
| Link lost                                 | Stay put or move to a better spot.                                                          |
| Low battery                               | Let it come home.                                                                           |
| Stuck in weed                             | Wait for up to three wiggles.                                                               |
| Boat stopped: position lost               | Wait up to a minute.                                                                        |
| At or beyond the fence                    | Let it return.                                                                              |
| Person or animal near the boat            | COME HOME, or STOP if close.                                                                |
| Boat dead in the water                    | It floats.                                                                                  |
| No internet                               | Choose a template with the crew..                                                           |
| Mission Control dead                      | The boat comes home by itself after 60 s.                                                   |
| Water or heat in the box                  | Let it return, key out, battery out, dry or cool, and investigate before the next mission.. |
| Boat behaving strangely                   | STOP.                                                                                       |
| Battery hot, swollen, smoking or smelling | Keep everyone away, especially the crew.                                                    |

## Appendix A. Traceability

| Requirement | Text (abridged)                                                                                                                                   | Met in                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| OPS-001     | The pre-launch checklist shall cover at least: site permission and conditions; no swim session, course or al...                                   | OP-04, OP-06, CL-05, CL-06, CL-08, CL-10, CL-13, CL-15 |
| OPS-002     | Written permission from Cambridge Sport Lakes Trust shall be obtained before any operation at Milton Country...                                   | OP-01, CL-01                                           |
| OPS-003     | There shall be no operation during swim sessions or watersports courses, or while a blue-green algae warning...                                   | OP-06, OP-12, CL-02                                    |
| OPS-004     | Exclusion zones shall keep the boat at least 20 m from occupied angling swims and other water users. If anyo...                                   | OP-01, OP-06, OP-10, OP-14, CL-03, K-06                |
| OPS-005     | Islands, reed beds and known nesting areas shall be exclusion zones, and every nest structure shall have an ...                                   | OP-01, OP-08, OP-10, OP-14, CL-04                      |
| OPS-006     | At Milton, the Mk1 inclusion fence shall stay within 100 m of the launch point until at least 5 successful l...                                   | OP-01                                                  |
| OPS-007     | An adult operator shall be present and in control throughout. The child shall never be unsupervised near the...                                   | OP-06, OP-07, OP-10, CL-17                             |
| OPS-008     | The recovery kit shall include a telescopic pole of at least 3 m with a landing net, and a casting rod with ...                                   | OP-04, OP-07, CL-15, K-07                              |
| OPS-009     | Trials shall progress: simulation, then bench, then paddling pool or small water, then lake. Each failsafe s...                                   | OP-02                                                  |
| OPS-010     | Batteries shall be charged under supervision in a fire-safe bag, and stored at storage voltage.                                                   | OP-04, OP-12, K-12                                     |
| OPS-011     | A spares kit shall be taken to every session: spare propellers, a spare fuse, cable ties, a microfibre lens ...                                   | OP-04, CL-16                                           |
| OPS-012     | Before the first lake trial, the operator shall rehearse every contingency scenario in section 3.5 in simula...                                   | OP-02, OP-03                                           |
| FMEA action | Action                                                                                                                                            | Met in                                                 |
| A-05        | Fence survey procedure: walk the launch bank with a GNSS reference to check imagery offset before the first mission at a site                     | OP-01                                                  |
| A-09        | Checklist and procedure: key out before anyone touches the boat; lift only by the handle or hoop; inspect guards                                  | OP-05, OP-06, OP-07, OP-09, OP-11, CL-09               |
| A-11        | Box temperature sensor with alarm and RTL above 60 °C; battery inspection before each session                                                     | OP-04, CL-07, K-10                                     |
| A-17        | Operator is the dedicated lookout; brief 'STOP if in doubt'; no missions while people are within 50 m of the planned route                        | OP-08, OP-10, CL-03, CL-17, K-06                       |
| A-19        | Site file lists the preferred launch point per wind direction; the checklist asks 'is the wind blowing towards us?' and suggests the launch point | OP-01, OP-06, CL-06                                    |